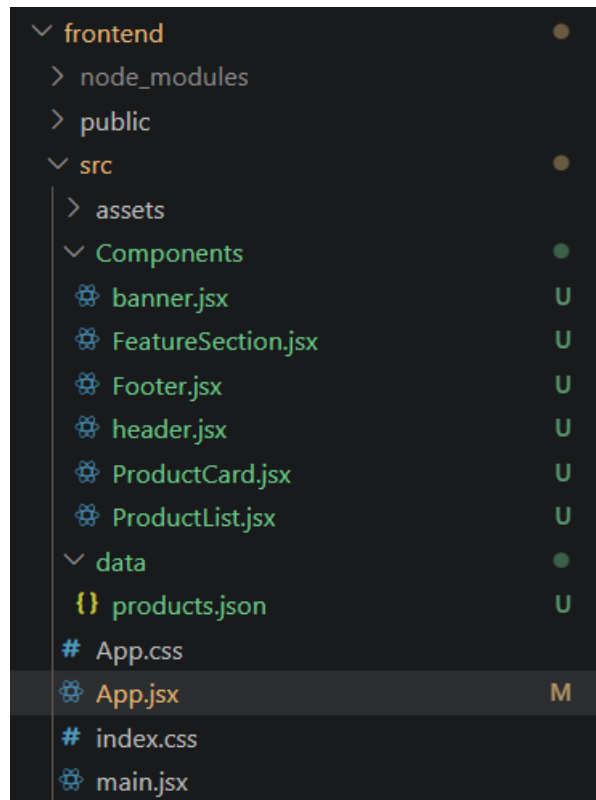


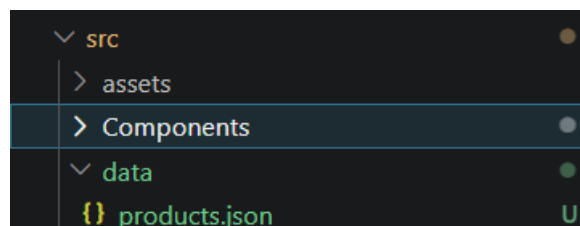
Session 3 – ShopHub Product Catalog

1. Starting Point

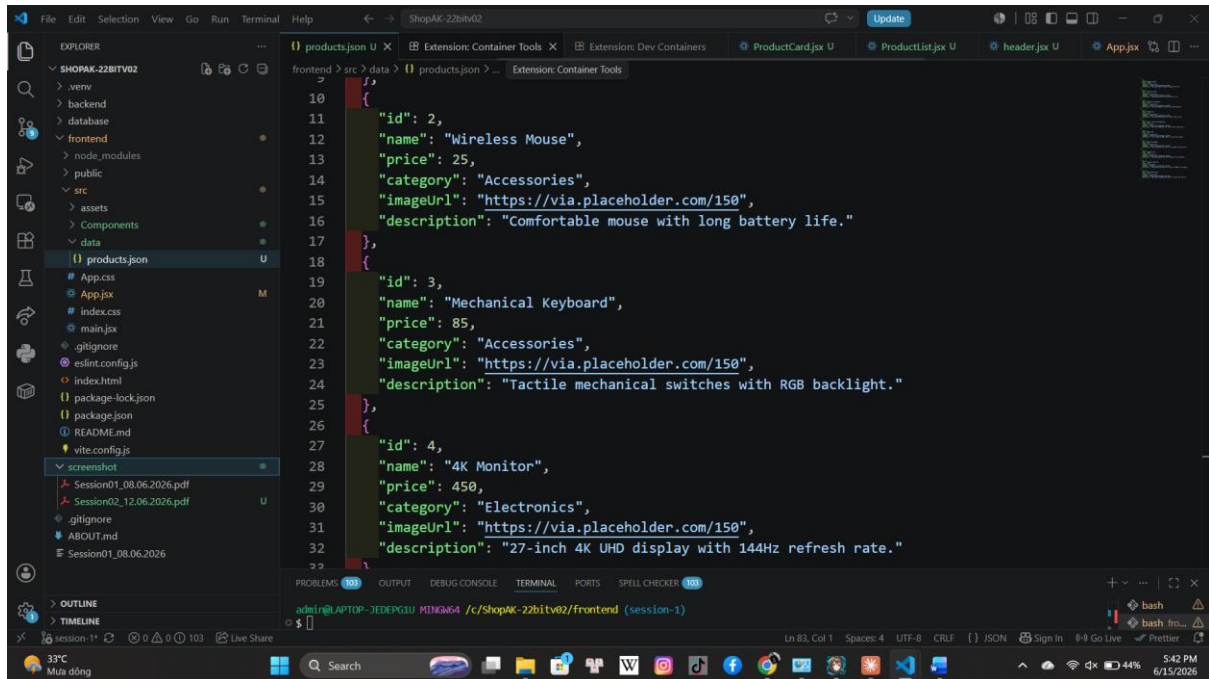


2. Create Mock Data

2.1. Create the data folder and file



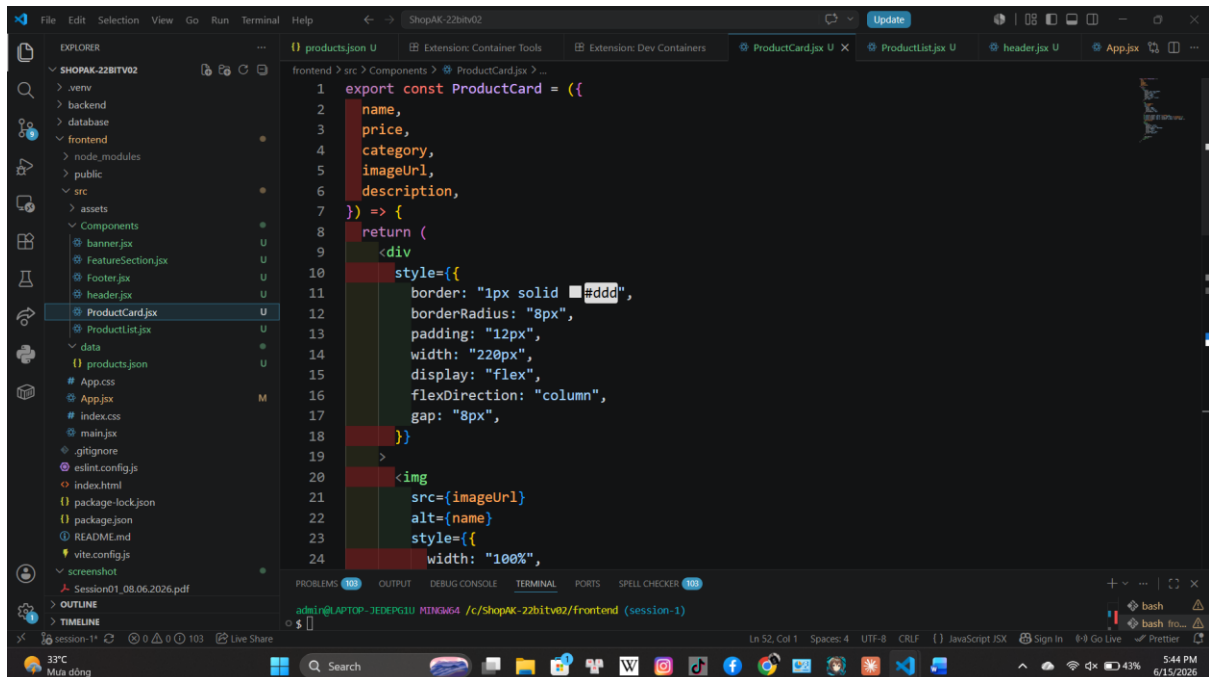
2.2. Add sample products



```
10  {
11    "id": 2,
12    "name": "Wireless Mouse",
13    "price": 25,
14    "category": "Accessories",
15    "imageUrl": "https://via.placeholder.com/150",
16    "description": "Comfortable mouse with long battery life."
17  },
18  {
19    "id": 3,
20    "name": "Mechanical Keyboard",
21    "price": 85,
22    "category": "Accessories",
23    "imageUrl": "https://via.placeholder.com/150",
24    "description": "Tactile mechanical switches with RGB backlight."
25  },
26  {
27    "id": 4,
28    "name": "4K Monitor",
29    "price": 450,
30    "category": "Electronics",
31    "imageUrl": "https://via.placeholder.com/150",
32    "description": "27-inch 4K UHD display with 144Hz refresh rate."
33  }
34 }
```

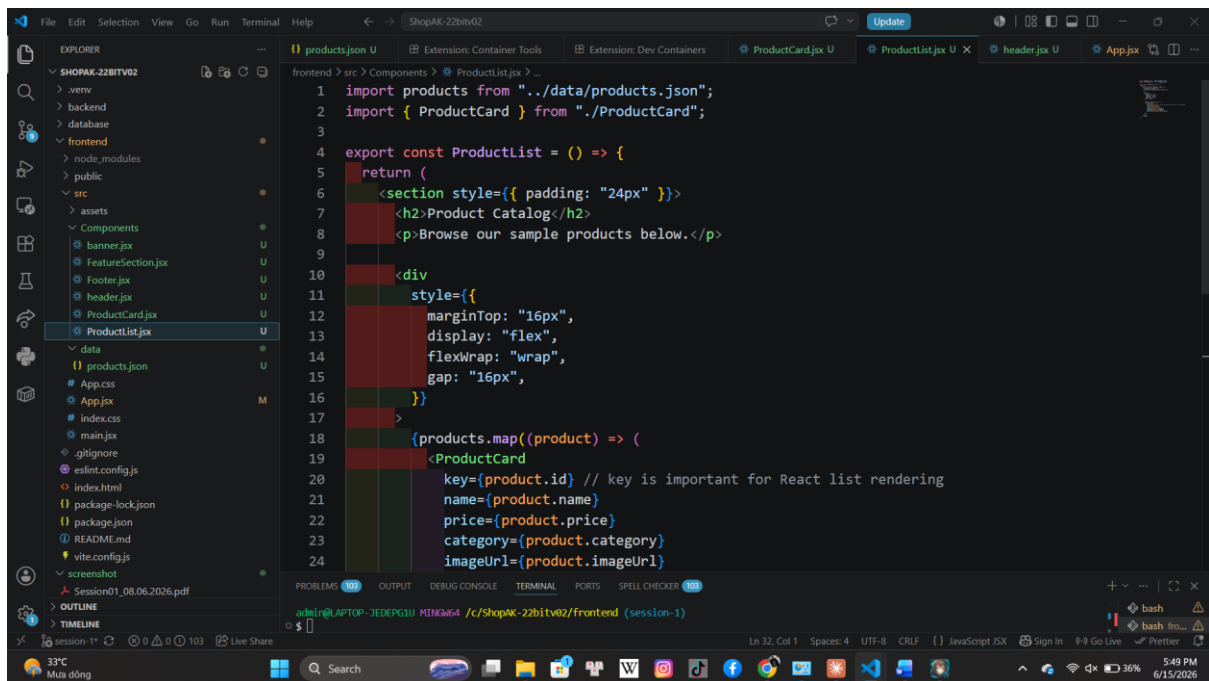
3. Create Productcard Component

3.1. Implementation



```
1  export const ProductCard = ({
2    name,
3    price,
4    category,
5    imageUrl,
6    description,
7  }) => {
8    return (
9      <div
10        style={{
11          border: "1px solid #ddd",
12          borderRadius: "8px",
13          padding: "12px",
14          width: "220px",
15          display: "flex",
16          flexDirection: "column",
17          gap: "8px",
18        }}
19      >
20        <img
21          src={imageUrl}
22          alt={name}
23          style={{
24            width: "100%",
```

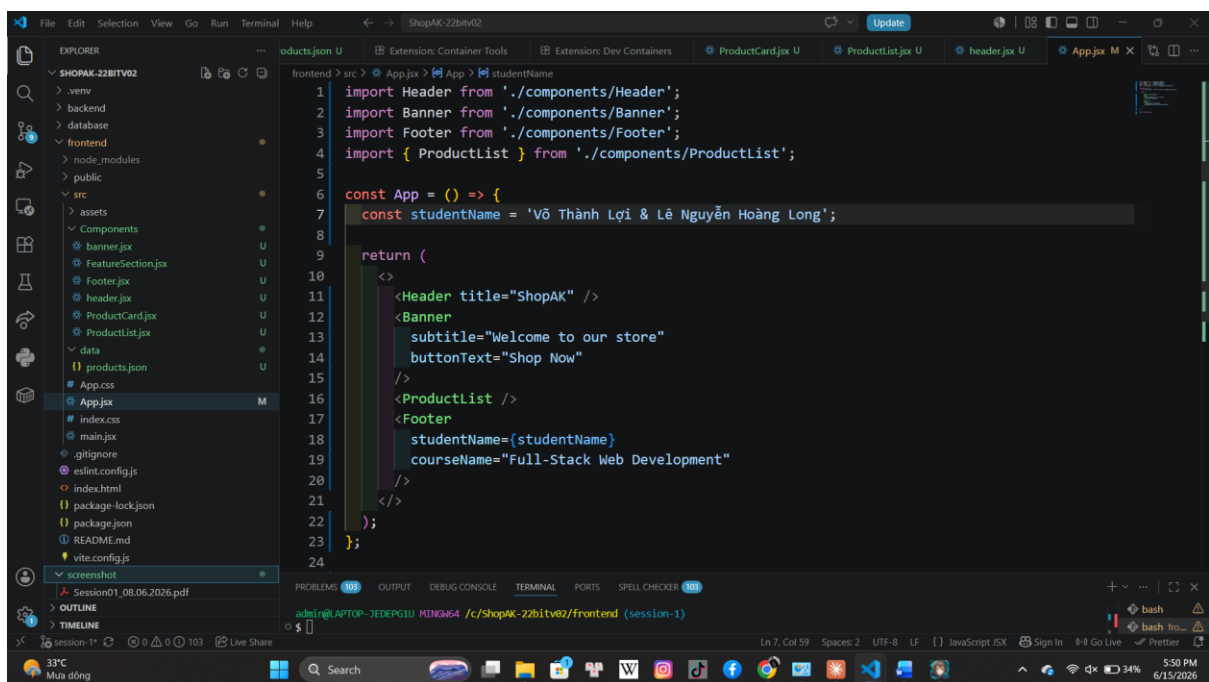
4. Create ProductList component with map()



The screenshot shows the VS Code editor with the `ProductList.js` file open. The file is located in the `src/components` directory. The code defines a `ProductList` component that uses the `map` function to render a list of products from the `products.json` file. The component uses the `ProductCard` component to display each product. The `App.js` file is also visible in the Explorer, showing the `App` component structure.

```
1 import products from '../data/products.json';
2 import { ProductCard } from './ProductCard';
3
4 export const ProductList = () => {
5   return (
6     <section style={{ padding: "24px" }}>
7       <h2>Product Catalog</h2>
8       <p>Browse our sample products below.</p>
9     <div
10       style={{
11         marginTop: "16px",
12         display: "flex",
13         flexWrap: "wrap",
14         gap: "16px",
15       }}
16     >
17       {products.map((product) => (
18         <ProductCard
19           key={product.id} // key is important for React list rendering
20           name={product.name}
21           price={product.price}
22           category={product.category}
23           imageUrl={product.imageUrl}
24         />
25       ))}
26     </div>
27   );
28 }
```

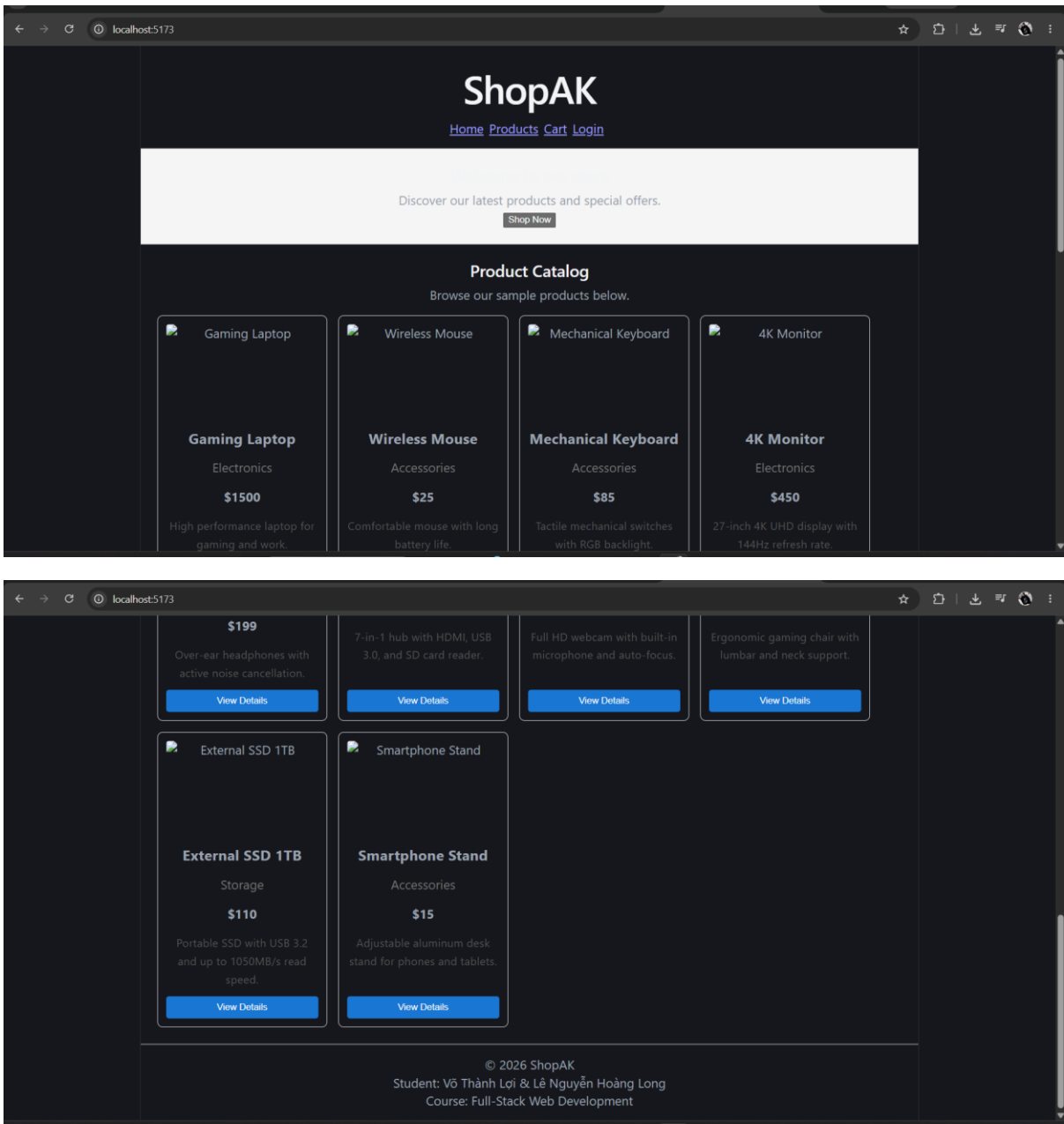
5. Integrate ProductList into the App



The screenshot shows the VS Code editor with the `App.js` file open. The file is located in the `src` directory. The code defines the `App` component, which imports the `Header`, `Banner`, `Footer`, and `ProductList` components. The `App` component uses the `ProductList` component to display the product catalog. The `App` component also includes a `Header` and a `Footer` component. The `App` component is rendered in the `App` component structure.

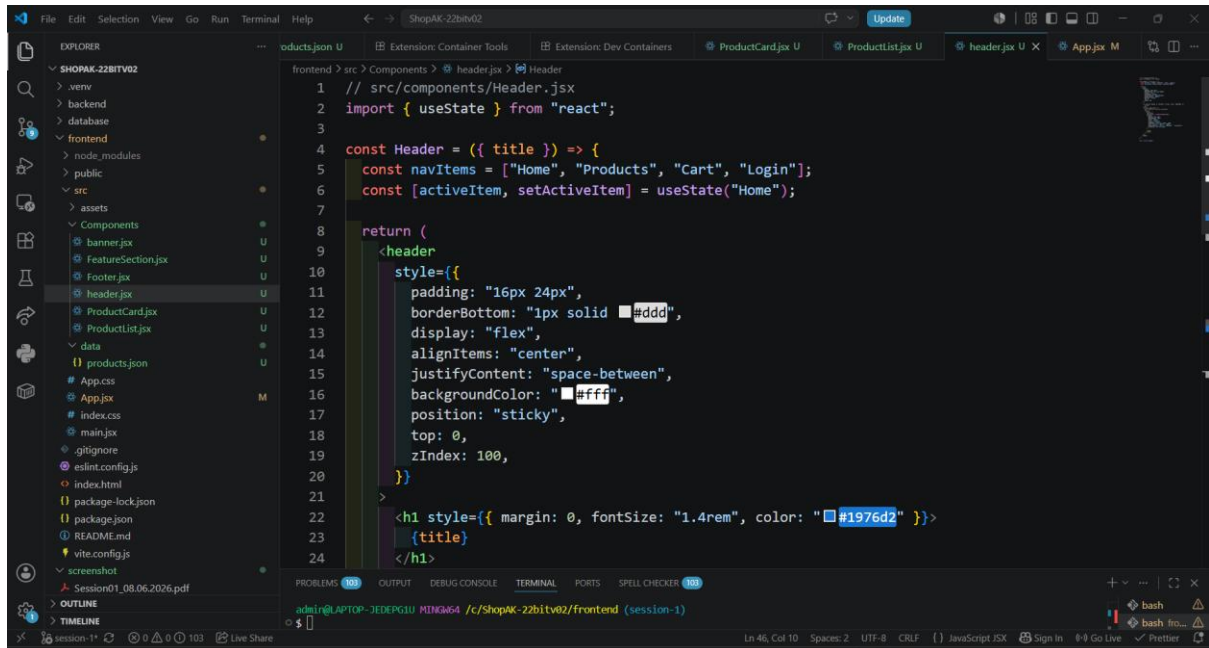
```
1 import Header from './components/Header';
2 import Banner from './components/Banner';
3 import Footer from './components/Footer';
4 import { ProductList } from './components/ProductList';
5
6 const App = () => {
7   const studentName = 'Võ Thành Lợi & Lê Nguyễn Hoàng Long';
8
9   return (
10     <>
11       <Header title="ShopAK" />
12       <Banner
13         subtitle="Welcome to our store"
14         buttonText="Shop Now"
15       />
16       <ProductList />
17       <Footer
18         studentName={studentName}
19         courseName="Full-Stack Web Development"
20       />
21     </>
22   );
23 };
24
```

Result



6. Navigation and Styling Enhancements

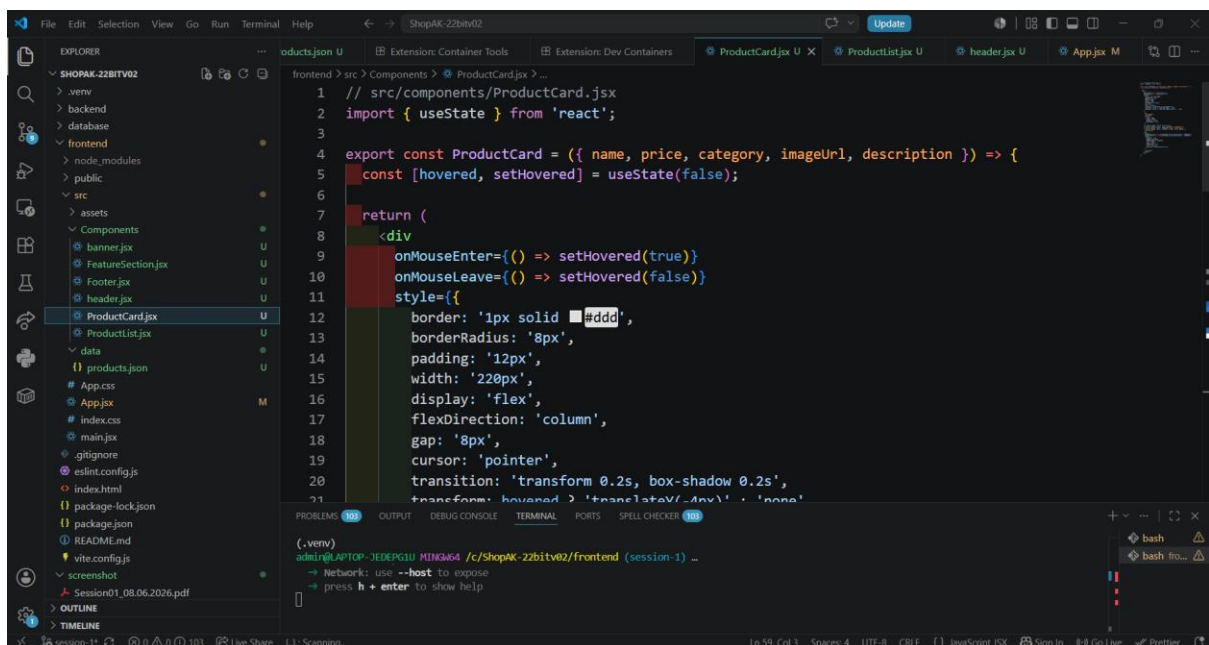
6.1. Navigation – Highlight “Products” in Header (basic version)



```
1 // src/components/Header.jsx
2 import { useState } from "react";
3
4 const Header = ({ title }) => {
5   const navItems = ["Home", "Products", "Cart", "Login"];
6   const [activeItem, setActiveItem] = useState("Home");
7
8   return (
9     <header
10       style={{
11         padding: "16px 24px",
12         borderBottom: "1px solid #ddd",
13         display: "flex",
14         alignItems: "center",
15         justifyContent: "space-between",
16         backgroundColor: "#fff",
17         position: "sticky",
18         top: 0,
19         zIndex: 100,
20       }}
21     >
22       <h1 style={{ margin: 0, fontSize: "1.4rem", color: "#1976d2" }}>
23         {title}
24       </h1>
25     </header>
26   );
27 }
```

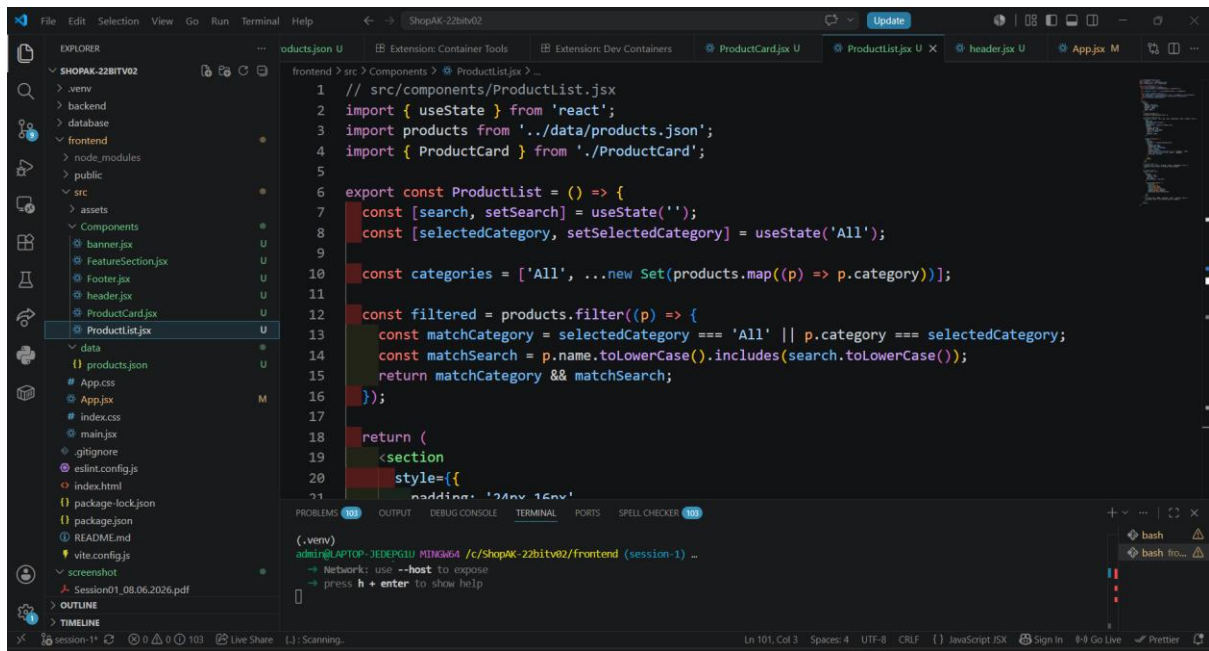
6.2. Basic Styling Improvements

ProductCard



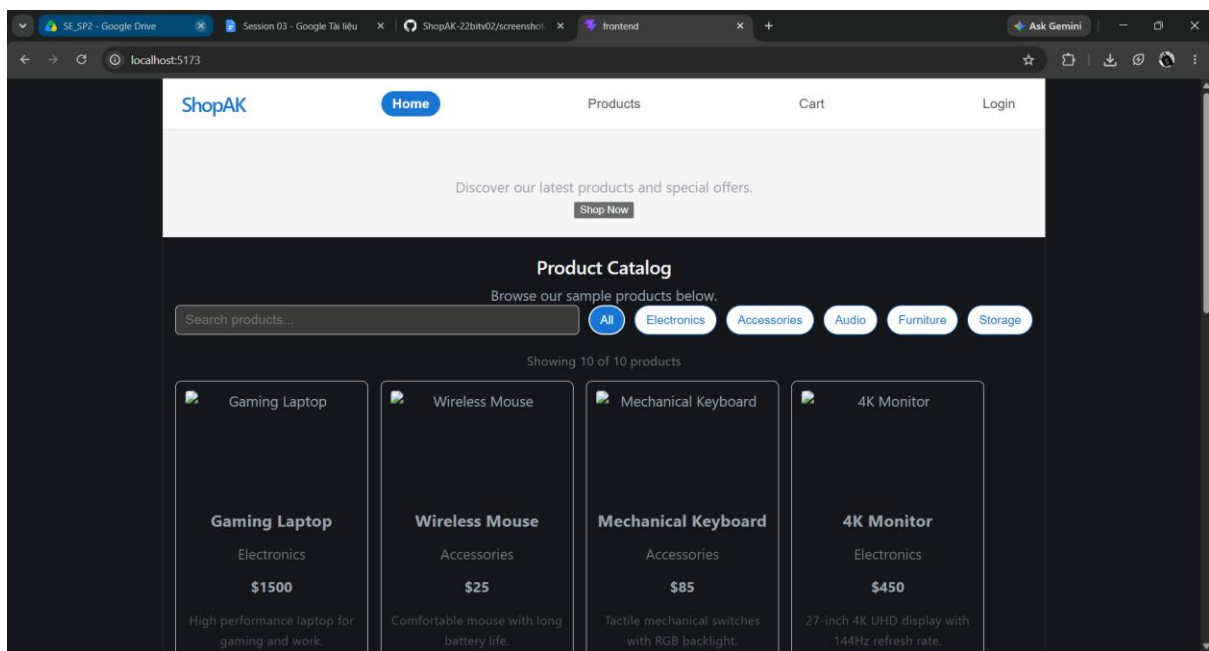
```
1 // src/components/ProductCard.jsx
2 import { useState } from "react";
3
4 export const ProductCard = ({ name, price, category, imageUrl, description }) => {
5   const [hovered, setHovered] = useState(false);
6
7   return (
8     <div
9       onMouseEnter={() => setHovered(true)}
10      onMouseLeave={() => setHovered(false)}
11      style={{
12        border: "1px solid #ddd",
13        borderRadius: "8px",
14        padding: "12px",
15        width: "220px",
16        display: "flex",
17        flexDirection: "column",
18        gap: "8px",
19        cursor: "pointer",
20        transition: "transform 0.2s, box-shadow 0.2s",
21        transform: hovered ? "translateY(-4px)" : "none"
22      }}
23     >
24       <img alt={imageUrl} />
25       <p>{name}</p>
26       <p>{price}</p>
27       <p>{category}</p>
28       <p>{description}</p>
29     </div>
30   );
31 }
```

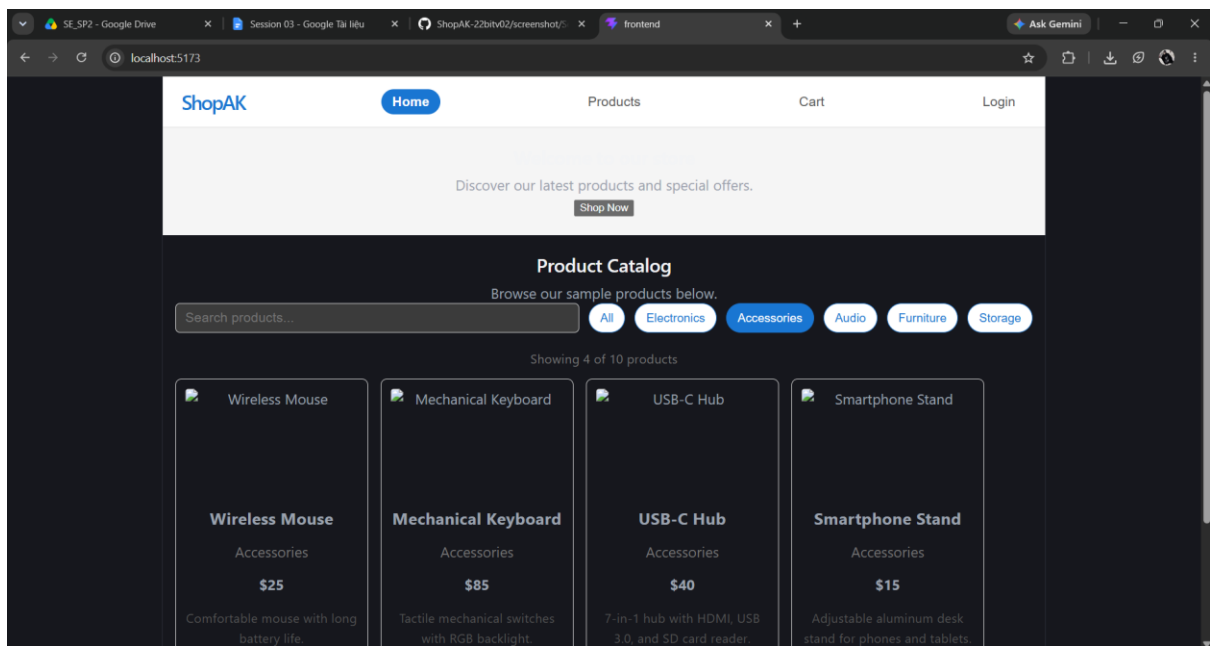
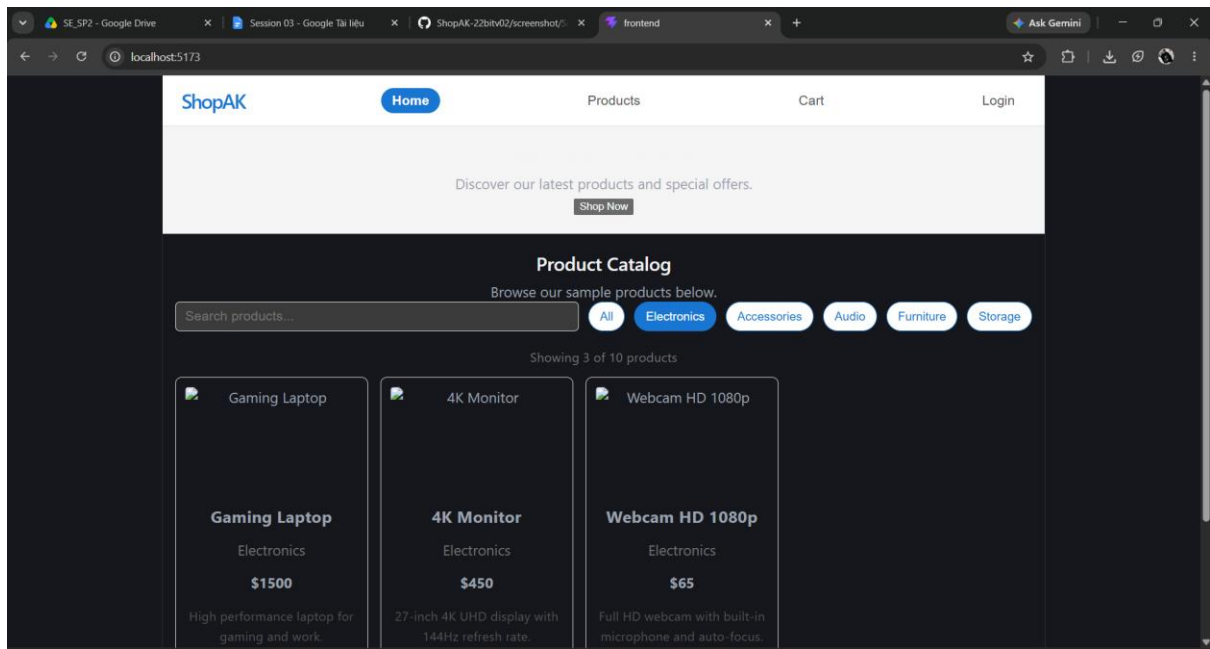
ProductList



```
1 // src/components/ProductList.jsx
2 import { useState } from 'react';
3 import products from '../data/products.json';
4 import { ProductCard } from './ProductCard';
5
6 export const ProductList = () => {
7   const [search, setSearch] = useState('');
8   const [selectedCategory, setSelectedCategory] = useState('All');
9
10  const categories = ['All', ...new Set(products.map((p) => p.category))];
11
12  const filtered = products.filter((p) => {
13    const matchCategory = selectedCategory === 'All' || p.category === selectedCategory;
14    const matchSearch = p.name.toLowerCase().includes(search.toLowerCase());
15    return matchCategory && matchSearch;
16  });
17
18  return (
19    <section
20      style={{
21        padding: '20px 16px'
22      }}
23    >
```

Result



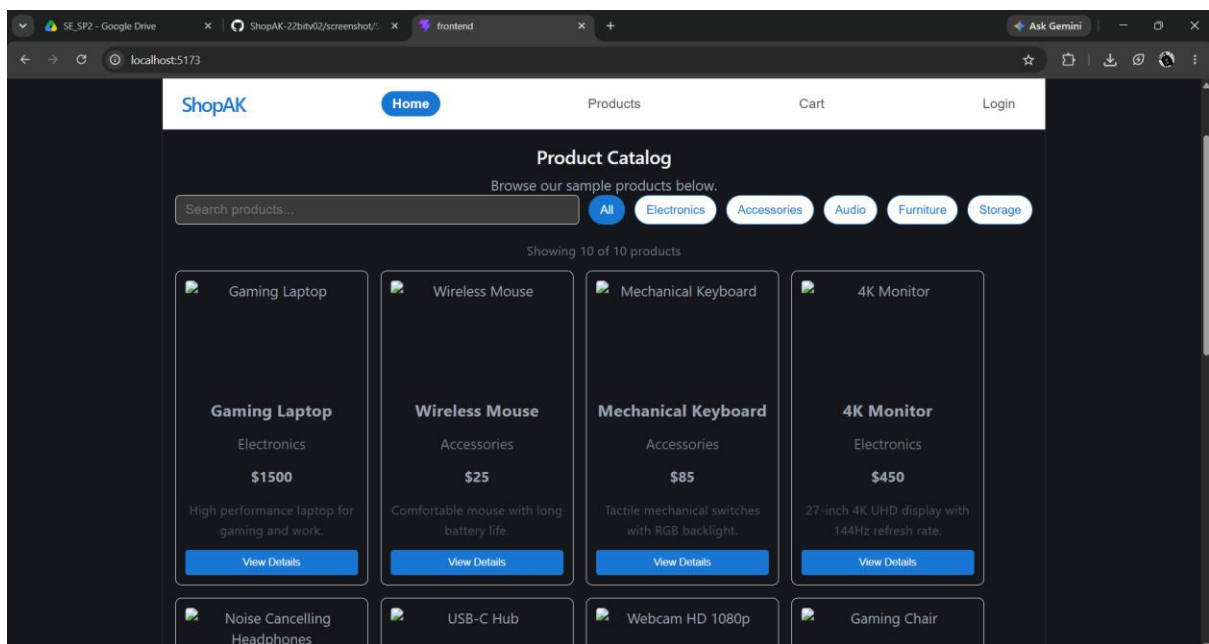


7. Stretch Tasks

```
frontend > src > Components > ProductCard.jsx > ...
1 // src/components/ProductCard.jsx
2 import { useState } from 'react';
3
4 export const ProductCard = ({ name, price, category, imageUrl, description }) => {
5   const [hovered, setHovered] = useState(false);
6
7   const truncatedDescription =
8     description.length > 70 ? description.slice(0, 70) + '...' : description;
9
10  return (
11    <div
12      onMouseEnter={() => setHovered(true)}
13      onMouseLeave={() => setHovered(false)}
14      style={{
15        border: '1px solid #ddd',
16        borderRadius: '8px',
17        padding: '12px',
18        width: '220px',
19        display: 'flex',
20        flexDirection: 'column',
21        gap: '8px'
22      }}
23    >
24      <img alt={imageUrl} data-bbox="125 125 155 155" data-kind="parent" data-rs="2" style="width: 100px; height: 100px; border-radius: 8px;"/>
25      <div>
26        <h3>{name}</h3>
27        <p>{category}</p>
28        <p>{truncatedDescription}</p>
29        <p>{price}</p>
30        <p>{description}</p>
31      </div>
32    </div>
33  );
34}
```

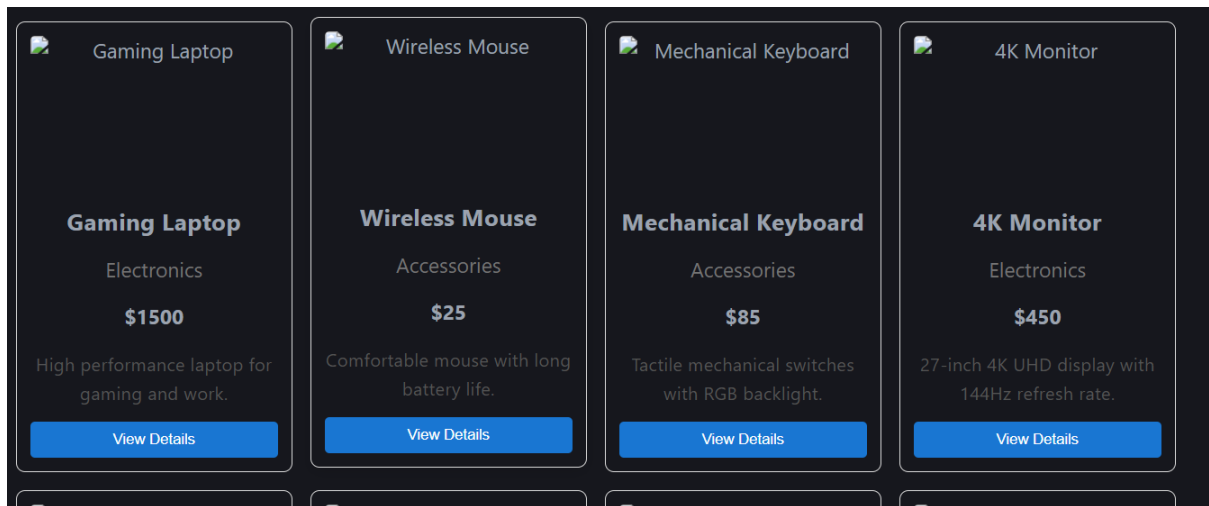
(.venv)
admin@LAPTOP-JEDEPG1U MINGW64 /c/ShopAK-22bitv02/frontend (session-1) ...
→ Network: use --host to expose
→ press h + enter to show help

Result



8. Expected Outcome

A ProductCard component that can render any product passed as props.

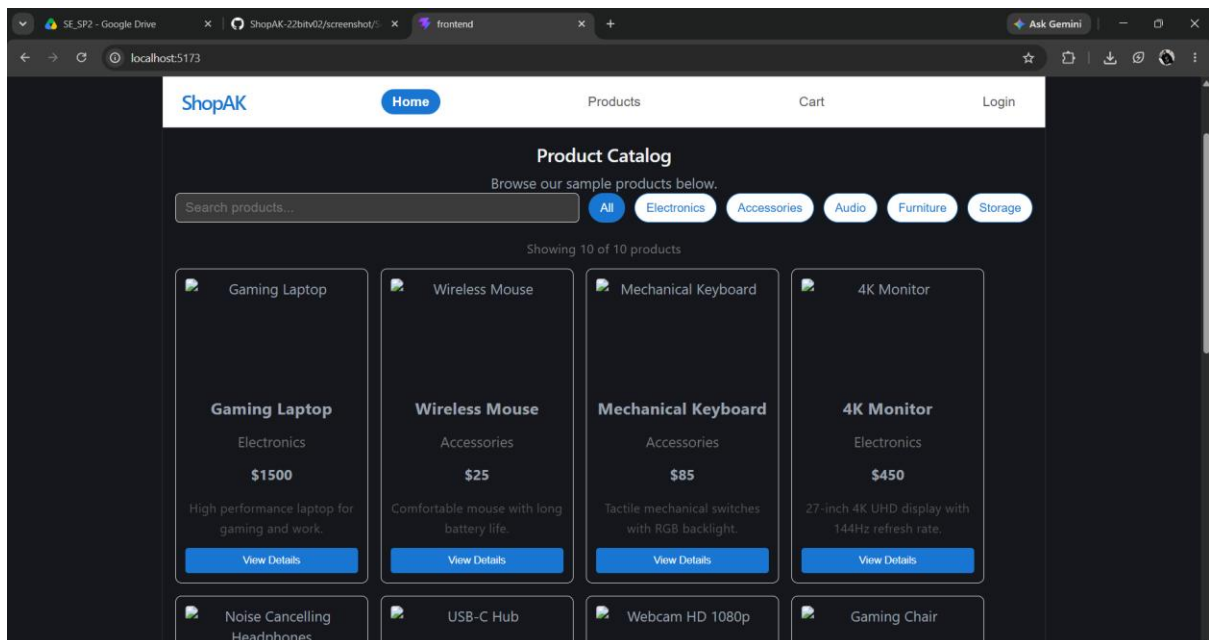


A **ProductList** component that uses **Array.map()** to render a list of **ProductCard** based on **products.json**.

```
const categories = ['All', ...new Set(products.map((p) => p.category))];

const filtered = products.filter((p) => {
  const matchCategory = selectedCategory === 'All' || p.category === selectedCategory;
  const matchSearch = p.name.toLowerCase().includes(search.toLowerCase());
  return matchCategory && matchSearch;
});
```

A **ShopHub** Home page displaying the product catalog in a card layout with basic navigation and styling.



Exercise

